Shree Rahul Education Society's (Regd.)
SHREE L. R. TIWARI COLLEGE OF ENGINEERING
Kanakia Park, Near Commissioner's Bungalow, Mira Road (East), Thane 401107, Maharashtra
(Approved by AICTE, Govt. of Maharashtra & Affiliated to University of Mumbai)
NAAC Accredited | ISO 9001:2015 Certified
Tel. No.: 022-28120144 / 022-28120145 | Email: slrtce@rahuleducation.com | Website: www.slrtce.in

DEPARTMENT OF COMPUTER ENGINEERING

CSL5605 Cloud Computing Lab

Sixth Semester, 2023-2024 (Even Semester)

Name of Student : Mainuddin Nayeem Khan

Roll No. : 53

Division : A

Batch :I

Day / Session : Monday

Venue : 308

Experiment No. :2

Title of Experiment : Introduction and Overview of Cloud Computing

Date of Conduction :

Date of Submission :

Particulars	Max. Marks	Marks Obtained
Preparedness and Efforts(PE)	**3**	
Knowledge of tools(KT)	**3**	
Debugging and results(DR)	**3**	
Documentation(DN)	**3**	
Punctuality & Lab Ethics(PL)	**3**	
Total	**15**	

Grades – Meet Expectations (3 Marks), Moderate Expectations (2 Marks), Below Expectations (1 Mark)

Checked and Verified by

Name of Faculty : Pravin Jangid

Signature :

Date :

Experiment 1

Aim:

Study of NIST model of Cloud Computing

Theory:

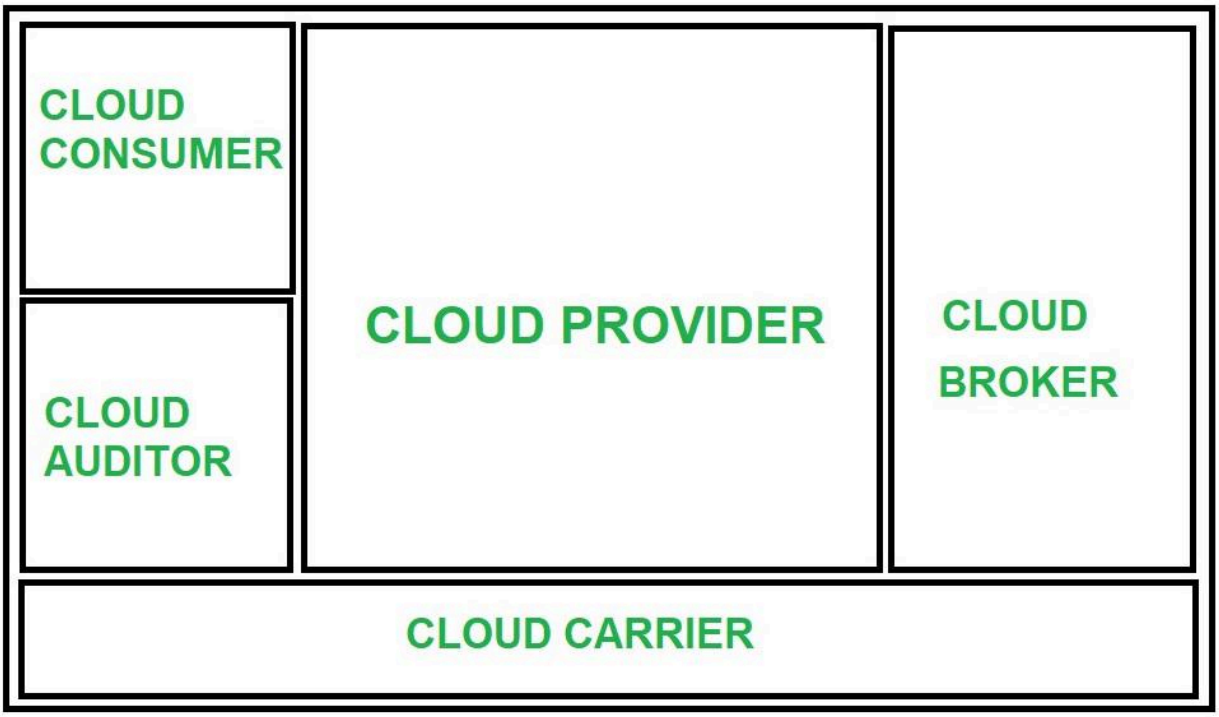
1.Define and explain cloud computing

Definition:
Cloud computing is a technology that provides computing resources such as servers, storage, software, and applications over the internet on a pay-per-use basis.

Explanation:
In cloud computing, data and applications are stored on remote servers instead of local computers. Users can access these resources anytime and from anywhere using the internet. It reduces the need for physical hardware and lowers maintenance cost.

Cloud computing offers features like scalability, flexibility, cost efficiency, and data backup. The main service models of cloud computing are IaaS, PaaS, and SaaS. Due to these benefits, cloud computing is widely used by organizations and individuals.

2.Draw and explain NIST cloud computing Architecture.



There are five major actors defined in the NIST cloud computing reference architecture.

- Cloud Provider – The entity that offers cloud services and manages the required infrastructure.
  It is responsible for service availability, security, and maintenance.

- Cloud Carrier – The intermediary that provides network connectivity between provider and consumer.
  It ensures reliable transport of cloud services.

- Cloud Broker – The party that manages and integrates services from multiple cloud providers.
  It helps consumers select and negotiate suitable cloud.

- Cloud Auditor – An independent entity that evaluates cloud services for security and compliance.
  It ensures standards and policies are being followed.

- Cloud Consumer – The user or organization that accesses and uses cloud services.
  It selects and utilizes services as per requirements.

3.Explain the need for cloud computing.

Need for Cloud Computing

- Cost Reduction – Cloud computing reduces the cost of purchasing hardware and software.
  Users pay only for the resources they use.

- Scalability – Resources can be easily increased or decreased based on demand.
  This helps organizations handle workload changes efficiently.

- Accessibility – Data and applications can be accessed anytime and from anywhere.
  It only requires an internet connection.

- Flexibility – Cloud services support different devices and operating systems.
  Users can work from multiple platforms easily.

- Data Backup and Recovery – Cloud provides automatic backup of data.
  It helps in quick recovery during system failures.

- High Availability – Cloud services ensure continuous access to applications.
  Downtime is minimized through reliable infrastructure.

4.Explain characteristics of cloud computing.

There are Characteristics of Cloud Computing

- On-Demand Self-Service – Users can automatically access computing resources whenever needed.
  No human interaction with the service provider is required.

- Broad Network Access – Cloud services are available over the internet.
  They can be accessed using devices like mobiles, laptops, and desktops.

- Resource Pooling – Computing resources are shared among multiple users.
  The provider dynamically allocates resources based on demand.

- Rapid Elasticity – Resources can be quickly scaled up or down.
  This allows flexibility according to workload requirements.

- Measured Service – Cloud usage is monitored and controlled.
  Users are charged only for the resources they consume.

5.Explain various service models of cloud computing with examples.

Service Models of Cloud Computing

Cloud computing provides services through different service models, each offering a different level of control and management. The main service models are:

- Infrastructure as a Service (IaaS)
  IaaS provides basic computing resources such as virtual machines, storage, and networking over the internet.
  Example: Amazon EC2, Microsoft Azure Virtual Machines.

- Platform as a Service (PaaS)
  PaaS provides a platform for developers to build, test, and deploy applications without managing underlying hardware.
  Example: Google App Engine, Heroku.

- Software as a Service (SaaS)
  SaaS provides ready-to-use software applications accessible through a web browser.
  Example: Gmail, Microsoft Office 365.

6.Explain various deployment models of cloud computing.

Deployment Models of Cloud Computing

Cloud computing deployment models describe how cloud services are made available to users. The main deployment models are:

- Public Cloud
  Public cloud services are provided by third-party vendors over the internet.
  These services are shared among multiple users and are cost-effective.

- Private Cloud
  Private cloud is used exclusively by a single organization.
  It offers better security and control over data.

- Hybrid Cloud
  Hybrid cloud is a combination of public and private clouds.
  It allows data and applications to be shared between both environments.

- Community Cloud
  Community cloud is shared by organizations with similar requirements.
  It is managed jointly or by a third party.

These deployment models help organizations choose the best cloud setup based on security, cost, and performance needs.

7.Advantages and disadvantages of cloud computing

Advantages:

- Reduces cost of hardware, software, and maintenance; pay only for what is used.
- Resources can be scaled up or down as per demand.
- Data and applications can be accessed anytime, anywhere with an internet connection.
- Provides automatic backup and disaster recovery, ensuring data safety.

Disadvantages:

- Requires a stable internet connection to access services.
- Data stored online may face security and privacy risks.
- Users have limited control over cloud infrastructure.
- Service outages or downtime can interrupt operations.

CONCLUSION

Which factors to consider before opting for Cloud Computing

Compare containers and Virtual Machine

Cloud computing is flexible, scalable, and cost-effective, allowing access to data and applications from anywhere. It reduces hardware needs and improves efficiency.

Factors to Consider:
Cost, security, performance, scalability, and vendor support before choosing a cloud service.

Containers vs Virtual Machines:

- Containers are lightweight, share the host OS, and start fast.
- VMs are heavier, include a full OS, and provide stronger isolation.
- Containers are portable and ideal for microservices; VMs are better for running multiple OS environments